

# Probability – III

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## References:

- Dimitri P. Bertsekas and John N. Tsitsiklis, *Introduction to Probability*, Chapter 2
- Walpole R. E., Myers R. H., Myers S. L. and Ye K., *Probability & Statistics for Engineers & Scientists*, Ch. 4
- William Navidi, *Statistics for Engineers and Scientists*, Ch. 2.6
- Slides are credited from Prof. Berlin Chen, NTNU.

# **JOINT DISTRIBUTED RANDOM VARIABLES**

# Motivation

- Given an experiment, e.g., a medical diagnosis
  - The results of blood test is modeled as numerical values of a random variable  $X$
  - The results of magnetic resonance imaging (MRI, 核磁共振攝影) is also modeled as numerical values of a random variable  $Y$

We would like to consider probabilities of events involving simultaneously the numerical values of these two variables and to investigate their mutual couplings

$$\mathbf{P}(\{X = x\} \cap \{Y = y\})?$$

# Joint Discrete Random Variables

- If  $X$  and  $Y$  are **jointly discrete** random variables:
  - The joint probability mass function of  $X$  and  $Y$  is the function

$$p_{X,Y}(x, y) = \mathbf{P}(X = x, Y = y)$$

- The marginal probability mass functions of  $X$  and  $Y$  can be obtained from the joint probability mass function as follows:

$$p_X(x) = \mathbf{P}(X = x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \mathbf{P}(Y = y) = \sum_x p_{X,Y}(x, y)$$

where the sums are taken over all the possible values of  $Y$  and of  $X$ , respectively

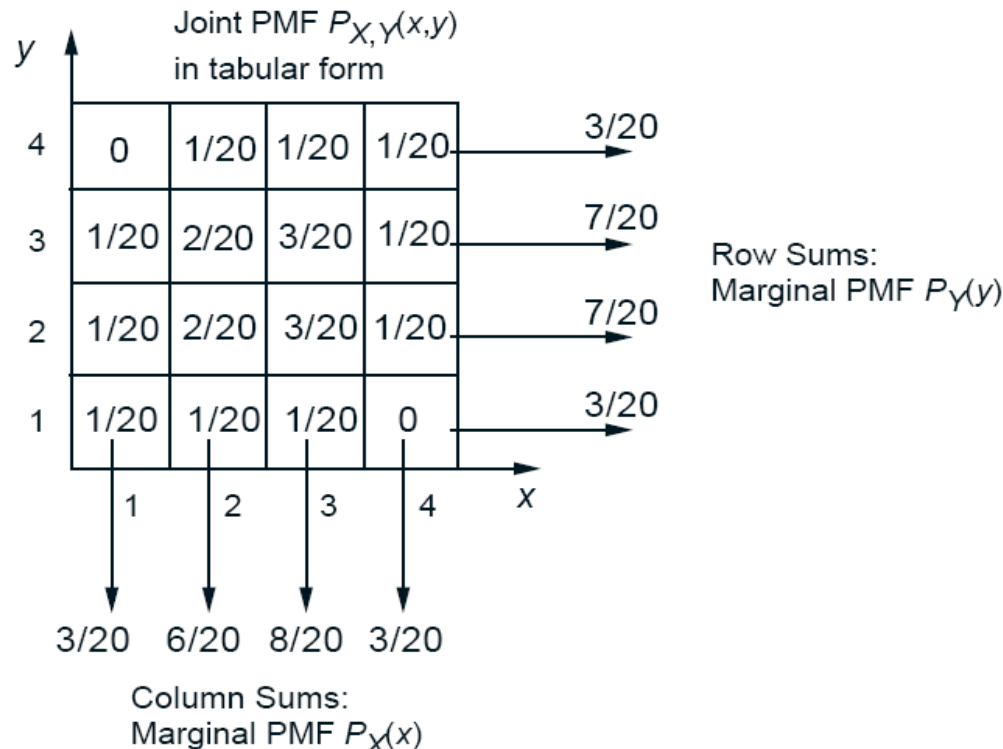
- The joint probability mass function has the property that

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

where the sum is taken over all the possible values of  $X$  and  $Y$

# Tabular Method

- Tabular Method:** Given the joint PMF of random variables  $X$  and  $Y$  is specified in a two-dimensional table, the marginal PMF of  $X$  and  $Y$  at a given value is obtained by adding the table entries along a corresponding column or row, respectively



# Example

- Example 2.52 , 2.53:

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**TABLE 2.3** Joint and marginal probability mass functions for the length  $X$  and width  $Y$  of a CD cover

| $x$      | $y$  |      | $p_X(x)$ |
|----------|------|------|----------|
|          | 15   | 16   |          |
| 129      | 0.12 | 0.08 | 0.20     |
| 130      | 0.42 | 0.28 | 0.70     |
| 131      | 0.06 | 0.04 | 0.10     |
| $p_Y(y)$ | 0.60 | 0.40 |          |

1. Find the probability that a CD cover has a length of 129 mm.

$$\begin{aligned}\mathbf{P}(X = 129) &= \mathbf{P}(X = 129, Y = 15) + \mathbf{P}(X = 129, Y = 16) \\ &= 0.12 + 0.08 = 0.20\end{aligned}$$

2. Find the probability that a CD cover has a width of 16 mm.

$$\begin{aligned}\mathbf{P}(Y = 16) &= \mathbf{P}(X = 129, Y = 16) + \mathbf{P}(X = 130, Y = 16) \\ &\quad + \mathbf{P}(X = 131, Y = 16) \\ &= 0.08 + 0.28 + 0.04 = 0.40\end{aligned}$$

# Example (1/2)

- Two ballpoint are selected at random from a box that contain 3 blue pens, 2 red pens, and 3 green pens. If  $X$  is the number of blue pens selected and  $Y$  is the number of red pens selected, find (a) the joint probability mass function  $p_{X,Y}(x,y)$  (b)  $\mathbf{P}_{X,Y}(\{x,y\} \in A)$ , where  $A$  is the region  $\{(x,y) \mid x + y \leq 1\}$

– Solution:

$$(a) \quad p_{X,Y}(x,y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{2-x-y}}{\binom{8}{2}}, \quad \text{for } x = 0,1,2; y = 0,1,2; \text{ and } 0 \leq x + y \leq 2$$

$$(b) \quad \begin{aligned} p_{X,Y}(x,y) &= \mathbf{P}(X + Y \leq 1) \\ &= p_{X,Y}(0,0) + p_{X,Y}(0,1) + p_{X,Y}(1,0) \\ &= \frac{3}{28} + \frac{3}{14} + \frac{9}{28} = \frac{9}{14} \end{aligned}$$

|               |   | $x$            |                 |                | Row Totals      |
|---------------|---|----------------|-----------------|----------------|-----------------|
|               |   | 0              | 1               | 2              |                 |
| $y$           | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
|               | 1 | $\frac{3}{14}$ | $\frac{3}{14}$  | 0              | $\frac{3}{7}$   |
|               | 2 | $\frac{1}{28}$ | 0               | 0              | $\frac{1}{28}$  |
| Column Totals |   | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1               |

# Example (2/2)

- Show that the column and row totals of Table give the marginal distribution of  $X$  alone and of  $Y$  alone.

$$\begin{aligned}
 p_X(x) &= \mathbf{P}(X = x) \\
 &= \sum_y \mathbf{P}(X = x, Y = y) \\
 &= \sum_y p_{X,Y}(x, y)
 \end{aligned}$$

|               |   | $x$            |                 |                | Row Totals      |
|---------------|---|----------------|-----------------|----------------|-----------------|
|               |   | 0              | 1               | 2              |                 |
| $y$           | 0 | $\frac{3}{28}$ | $\frac{9}{28}$  | $\frac{3}{28}$ | $\frac{15}{28}$ |
|               | 1 | $\frac{3}{14}$ | $\frac{3}{14}$  | 0              | $\frac{3}{7}$   |
|               | 2 | $\frac{1}{28}$ | 0               | 0              | $\frac{1}{28}$  |
| Column Totals |   | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1               |

– Solution:

$$\mathbf{P}(X = 0) = p_{X,Y}(0,0) + p_{X,Y}(0,1) + p_{X,Y}(0,2)$$

$$= \frac{3}{28} + \frac{3}{14} + \frac{1}{28} = \frac{5}{14}$$

$$\mathbf{P}(X = 1) = p_{X,Y}(1,0) + p_{X,Y}(1,1) + p_{X,Y}(1,2)$$

$$= \frac{9}{28} + \frac{3}{14} + 0 = \frac{15}{28}$$

$$\mathbf{P}(X = 2) = p_{X,Y}(2,0) + p_{X,Y}(2,1) + p_{X,Y}(2,2)$$

$$= \frac{3}{28} + 0 + 0 = \frac{3}{28}$$

$$\mathbf{P}(Y = 0) = p_{X,Y}(0,0) + p_{X,Y}(1,0) + p_{X,Y}(2,0)$$

$$= \frac{3}{28} + \frac{9}{28} + \frac{3}{28} = \frac{15}{28}$$

$$\mathbf{P}(Y = 1) = p_{X,Y}(0,1) + p_{X,Y}(1,1) + p_{X,Y}(2,1)$$

$$= \frac{3}{14} + \frac{3}{14} + 0 = \frac{6}{14}$$

$$\mathbf{P}(Y = 2) = p_{X,Y}(0,2) + p_{X,Y}(1,2) + p_{X,Y}(2,2)$$

$$= \frac{1}{28} + 0 + 0 = \frac{1}{28}$$



# Joint Continuous Random Variables

- If  $X$  and  $Y$  are **jointly continuous** random variables, with joint probability density function  $f_{X,Y}(x,y)$ , and  $a < b$ ,  $c < d$ , then

$$\mathbf{P}(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f_{X,Y}(x, y) dy dx$$

- The joint probability density function has the property that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy dx = 1$$

# Marginals of $X$ and $Y$

- If  $X$  and  $Y$  are **jointly continuous** with joint probability density function  $f_{X,Y}(x,y)$ , then the **marginal** probability density functions of  $X$  and  $Y$  are given, respectively, by

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx$$

- Such a process is called “marginalization”

# Example

- Example 2.54: Assume that for a certain type of washer, both the thickness and the hole diameter vary from item to item. Let  $X$  denote the thickness in millimeters and let  $Y$  denote the hole diameter in millimeters, for a randomly chosen washer. Assume the joint PDF of  $X$  and  $Y$  is given by

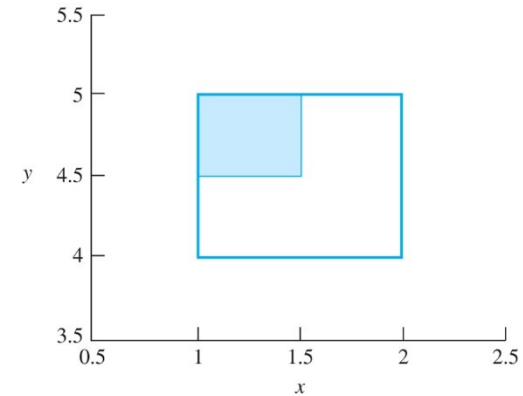
$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{6}(x+y) & \text{if } 1 \leq x \leq 2 \text{ and } 4 \leq y \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

Find the probability that a randomly chosen washer has a thickness between 1.0 and 1.5 mm, and a hole diameter between 4.5 and 5 mm.

# Answer

- Example 2.54:

We integrate the joint probability density function over the indicated region:



$$\begin{aligned}\mathbf{P}(1 \leq X \leq 1.5, 4.5 \leq Y \leq 5) &= \int_1^{1.5} \int_{4.5}^5 \frac{1}{6}(x+y) dy dx \\ &= \int_1^{1.5} \left( \frac{xy}{6} + \frac{y^2}{12} \Big|_{4.5}^5 \right) dx \\ &= \int_1^{1.5} \left( \frac{x}{12} + \frac{19}{48} \right) dx \\ &= \frac{1}{4}\end{aligned}$$

# What are the marginal PDFs of $X$ and $Y$ ?

- Example 2.54:

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{6}(x+y) & \text{if } 1 \leq x \leq 2 \text{ and } 4 \leq y \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x,y) dy \\ &= \int_4^5 \frac{1}{6}(x+y) dy \\ &= \frac{1}{6} \left( x + \frac{9}{2} \right) \quad \text{for } 1 \leq x \leq 2 \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f(x,y) dx \\ &= \int_1^2 \frac{1}{6}(x+y) dx \\ &= \frac{1}{6} \left( y + \frac{3}{2} \right) \quad \text{for } 4 \leq y \leq 5 \end{aligned}$$

# More than Two Random Variables

- If the random variables  $X_1, \dots, X_n$  are **jointly discrete**, the joint probability mass function is

$$p(x_1, \dots, x_n) = \mathbf{P}(X_1 = x_1, \dots, X_n = x_n)$$

- If the random variables  $X_1, \dots, X_n$  are **jointly continuous**, they have a joint probability density function  $f(x_1, x_2, \dots, x_n)$ , where

$$\mathbf{P}(a_1 \leq X_1 \leq b_1, \dots, a_n \leq X_n \leq b_n) = \int_{a_n}^{b_n} \cdots \int_{a_1}^{b_1} f(x_1, \dots, x_n) dx_1 \dots dx_n$$

for any constants  $a_1 \leq b_1, \dots, a_n \leq b_n$

# Means of Functions of Random Variables

- Let  $X$  be a random variable, and let  $h(X)$  be a function of  $X$ .

Then:

- If  $X$  is a **discrete** with probability mass function  $p(x)$ , then mean of  $h(X)$  is given by

$$\mu_{h(X)} = \sum_x h(x)p(x) \quad , \text{also denoted as } \mathbf{E}[h(X)]$$

where the sum is taken over all the possible values of  $X$

- If  $X$  is **continuous** with probability density function  $f(x)$ , the mean of  $h(x)$  is given by

$$\mu_{h(X)} = \int_{-\infty}^{\infty} h(x)f(x)dx \quad , \text{also denoted as } \mathbf{E}[h(X)]$$

# Functions of Joint Random Variables

- If  $X$  and  $Y$  are jointly distributed random variables, and  $h(X,Y)$  is a function of  $X$  and  $Y$ , then
  - If  $X$  and  $Y$  are jointly **discrete** with joint probability mass function  $p(x,y)$ ,

$$\mu_{h(X,Y)} = \sum_x \sum_y h(x,y)p(x,y)$$

where the sum is taken over all possible values of  $X$  and  $Y$

- If  $X$  and  $Y$  are jointly **continuous** with joint probability density function  $f(x,y)$ ,

$$\mu_{h(X,Y)} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x,y)f(x,y)dxdy$$



# Example

- Given the random variables  $X$  and  $Y$  whose joint is given in the following figure, and a new random variable  $Z$  is defined by  $Z = X + 2Y$ , calculate  $E[Z]$

- Method 1: (using linearity)

$$E[X] = 1 \cdot \frac{3}{20} + 2 \cdot \frac{6}{20} + 3 \cdot \frac{8}{20} + 4 \cdot \frac{3}{20} = \frac{51}{20}$$

$$E[Y] = 1 \cdot \frac{3}{20} + 2 \cdot \frac{7}{20} + 3 \cdot \frac{7}{20} + 4 \cdot \frac{3}{20} = \frac{50}{20}$$

$$E[Z] = E[X] + 2E[Y] = \frac{51}{20} + 2 \cdot \frac{50}{20} = \frac{151}{20} = 7.55$$

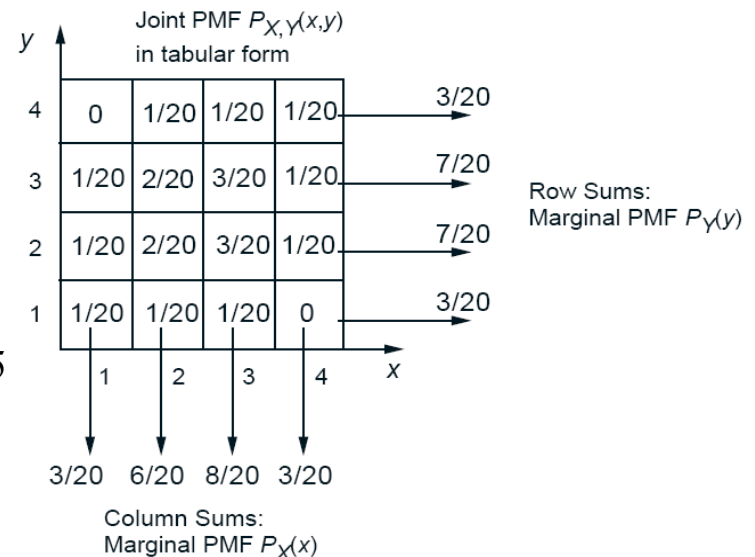
- Method 2:

$$p_Z(z) = \sum_{\{(x,y) | x+2y=z\}} p_{X,Y}(x,y)$$

$$p_Z(3) = \frac{1}{20}, p_Z(4) = \frac{1}{20}, p_Z(5) = \frac{2}{20}, p_Z(6) = \frac{2}{20}$$

$$p_Z(7) = \frac{4}{20}, p_Z(8) = \frac{3}{20}, p_Z(9) = \frac{3}{20}, p_Z(10) = \frac{2}{20}$$

$$p_Z(11) = \frac{1}{20}, p_Z(12) = \frac{1}{20}$$



$$\begin{aligned} \therefore E[Z] &= 3 \cdot \frac{1}{20} + 4 \cdot \frac{1}{20} + 5 \cdot \frac{2}{20} + 6 \cdot \frac{2}{20} \\ &\quad + 7 \cdot \frac{4}{20} + 8 \cdot \frac{3}{20} + 9 \cdot \frac{3}{20} + 10 \cdot \frac{2}{20} \\ &\quad + 11 \cdot \frac{1}{20} + 12 \cdot \frac{1}{20} = 7.55 \end{aligned}$$

# Example

- Example 2.56: The article “Performance Comparison of Two Location Based Routing Protocols for Ad Hoc Networks” (T. Camp, J. Boleng, et al., *Proceedings of the Twenty-first Annual Joint Conference of IEEE Computer and Communications Societies* 2002:1678-1687) describes a model for the movement of a mobile computer. Assume that a mobile computer moves within the region  $A$  bounded by the  $x$  axis, the line  $x = 1$ , and the line  $y = x$  in such a way that if  $(X, Y)$  denotes the position of the computer at a given time, the joint density of  $X$  and  $Y$  is given by

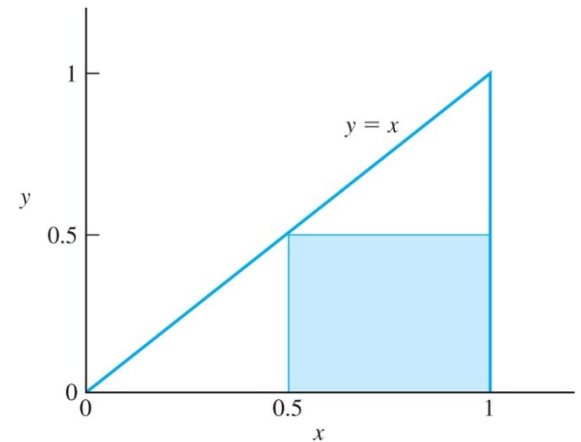
$$f_{X,Y}(x, y) = \begin{cases} 8xy & (x, y) \in A \\ 0 & (x, y) \notin A \end{cases}$$

Find  $\mathbf{P}(X > 0.5 \text{ and } Y < 0.5)$ .

# Answer

- Example 2.56:

We integrate the joint probability density function over the indicated region:



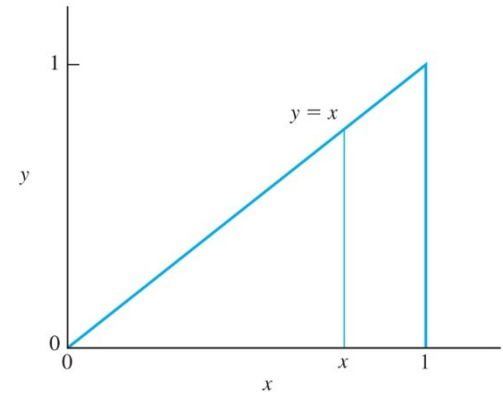
$$\begin{aligned}\mathbf{P}(X > 0.5 \text{ and } Y < 0.5) &= \int_{0.5}^1 \int_0^{0.5} 8xy \, dy \, dx \\ &= \int_{0.5}^1 \left\{ 4xy^2 \Big|_{y=0}^{y=0.5} \right\} dx \\ &= \int_{0.5}^1 x \, dx \\ &= 0.375\end{aligned}$$

# Answer

- Example 2.56:

Find the **marginal density of  $X$** .

To compute  $f_X(x)$ , the marginal density of  $X$ , we fix  $x$  and integrate the joint density along the vertical line through  $x$ , as shown in Figure. The integration is with respect to  $y$ , and the limitations are  $y = 0$  to  $y = x$ .



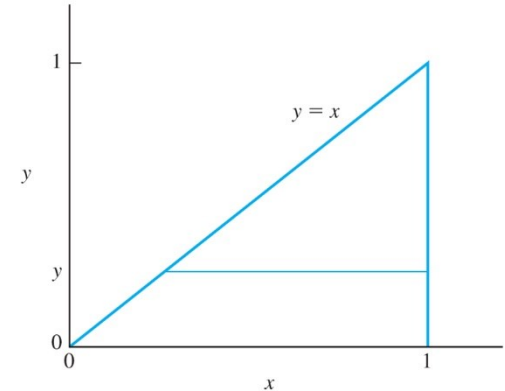
$$\begin{aligned} f_X(x) &= \int_0^x 8xy \, dy \\ &= 4xy^2 \Big|_{y=0}^{y=x} \\ &= 4x^3 \quad \text{for } 0 < x < 1 \end{aligned}$$

# Answer

- Example 2.56:

Find the **marginal density of  $Y$** .

To compute  $f_Y(y)$ , the marginal density of  $Y$ , we fix  $y$  and integrate the joint density along the horizontal line through  $y$ , as shown in Figure. The integration is with respect to  $x$ , and the limitations are  $x = y$  to  $x = 1$ .



$$\begin{aligned} f_Y(y) &= \int_y^1 8xy dx \\ &= 4x^2 y \Big|_{x=y}^{x=1} \\ &= 4y - 4y^3 \quad \text{for } 0 < y < 1 \end{aligned}$$

# Conditioning

- Recall that conditional probability provides us with a way to reason about the outcome of an experiment, based on partial information
- In the same spirit, latter, we can define **conditional distribution function (a PMF or a PDF)**, given the occurrence of a certain event or given the value of another random variable

# Conditioning a Random Variable on an Event (1/2)

- The **conditional PMF** of a random variable  $X$ , conditioned on a particular event  $A$  with  $\mathbf{P}(A) > 0$ , is defined by (where  $X$  and  $A$  are associated with the same experiment)

$$p_{X|A}(x) = \mathbf{P}(X = x|A) = \frac{\mathbf{P}(\{X = x\} \cap A)}{\mathbf{P}(A)}$$

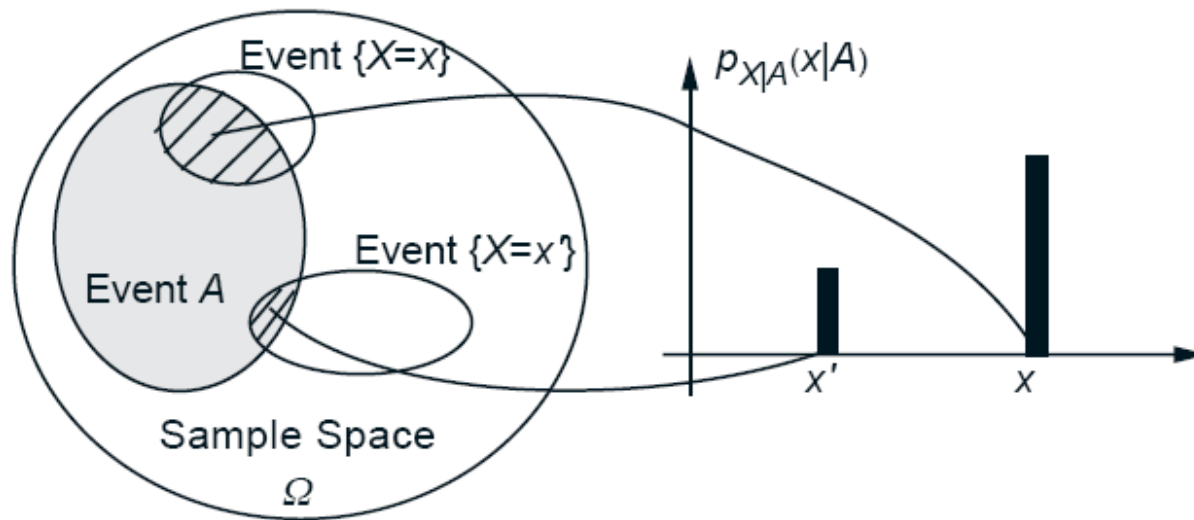
- Normalization Property
  - Note that the events  $\mathbf{P}(\{X = x\} \cap A)$  are **disjoint** for different values of  $X$ , their union is  $A$

$$\mathbf{P}(A) = \sum_x \mathbf{P}(\{X = x\} \cap A) \quad \text{Total probability theorem}$$

$$\therefore \sum_x p_{X|A}(x) = \sum_x \frac{\mathbf{P}(\{X = x\} \cap A)}{\mathbf{P}(A)} = \frac{\sum_x \mathbf{P}(\{X = x\} \cap A)}{\mathbf{P}(A)} = \frac{\mathbf{P}(A)}{\mathbf{P}(A)} = 1$$

# Conditioning a Random Variable on an Event (2/2)

- A graphical illustration



**Figure 2.12:** Visualization and calculation of the conditional PMF  $p_{X|A}(x)$ . For each  $x$ , we add the probabilities of the outcomes in the intersection  $\{X = x\} \cap A$  and normalize by dividing with  $\mathbf{P}(A)$ .

$p_{X|A}(x)$  is obtained by adding the probabilities of the outcomes that give rise to  $X = x$  and belong to the conditioning event  $A$



# Example

- **Example 2.12.** Let  $X$  be the roll of a fair six-sided die and  $A$  be the event that the roll is an even number

$$\begin{aligned} p_{X|A}(x) &= \mathbf{P}(X = x | \text{roll is even}) \\ &= \frac{\mathbf{P}(X = x \text{ and } X \text{ is even})}{\mathbf{P}(X \text{ is even})} \\ &\begin{cases} 1/3, & \text{if } x = 2, 4, 6 \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

# Discrete Conditional Distributions (1/2)

- Let  $X$  and  $Y$  be jointly **discrete** random variables, with joint probability density function  $p_{X,Y}(x,y)$ , let  $p_X(x)$  denote the marginal probability mass function of  $X$  and let  $x$  be any number for which  $p_X(x) > 0$ .

- The conditional probability mass function of  $Y$  given  $X = x$  is

$$\begin{aligned} p_{Y|X}(y|x) &= \mathbf{P}(Y = y | X = x) = \frac{\mathbf{P}(X = x, Y = y)}{\mathbf{P}(X = x)} \\ &= \frac{p_{X,Y}(x, y)}{p_X(x)} \end{aligned}$$

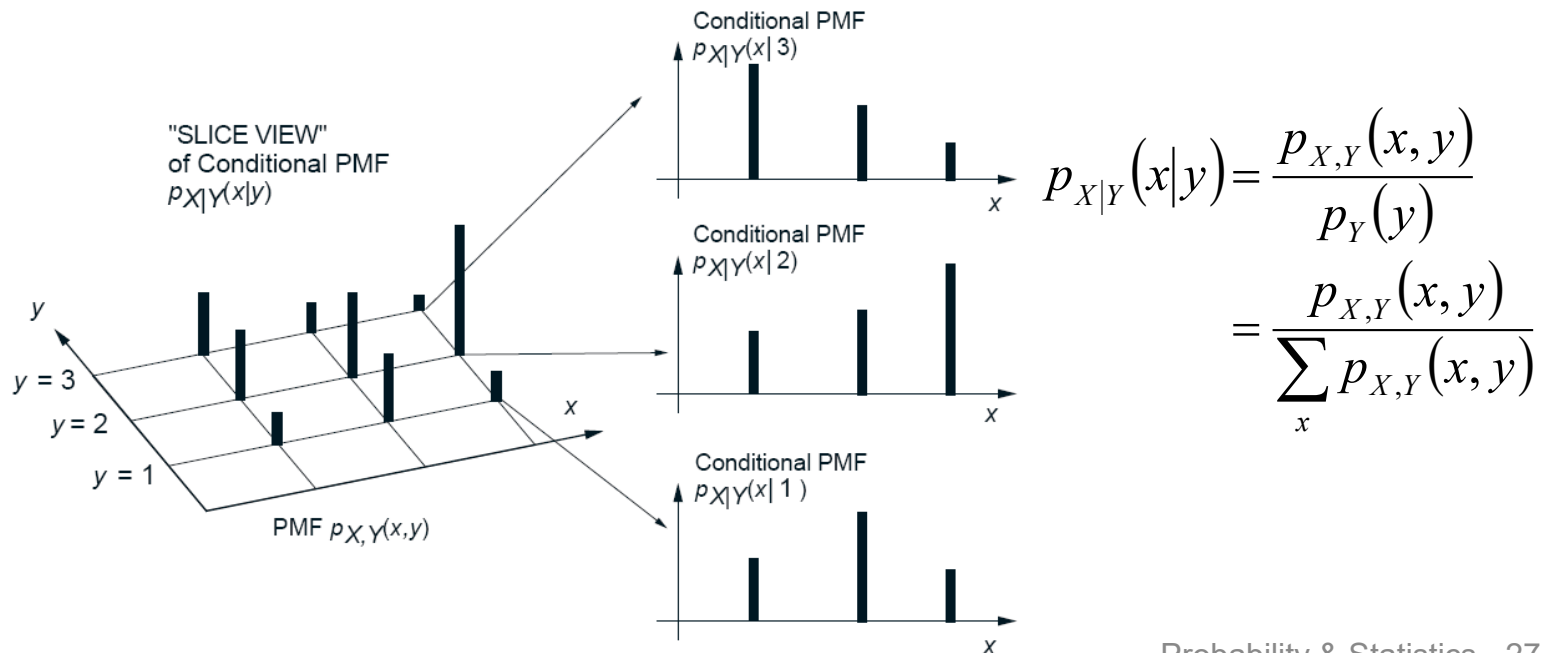
- Note that for any particular values of  $x$  and  $y$ , the value of  $p_{Y|X}(y|x)$  is just the conditional probability  $\mathbf{P}(Y=y | X=x)$

# Discrete Conditional Distributions (2/2)

- The conditional PMF can also be used to calculate the marginal PMFs

$$p_X(x) = \sum_y p_{X,Y}(x,y) = \sum_y p_Y(y)p_{X|Y}(x|y)$$

- Visualization of the conditional PMF  $p_{X|Y}$



# Continuing Example 2.52, 2.53

- Compute the conditional probability mass function  $p_{Y|X}(y|130)$ .

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**TABLE 2.3** Joint and marginal probability mass functions for the length  $X$  and width  $Y$  of a CD cover

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|----------|------|------|----------|
|          | 15   | 16   |          |
| 129      | 0.12 | 0.08 | 0.20     |
| 130      | 0.42 | 0.28 | 0.70     |
| 131      | 0.06 | 0.04 | 0.10     |
| $p_Y(y)$ | 0.60 | 0.40 |          |

The possible values for  $Y$  are  $y = 15$  and  $y = 16$ .

$$\begin{aligned}
 p_{Y|X}(15|130) &= \mathbf{P}(Y = 15 | X = 130) \\
 &= \frac{\mathbf{P}(Y = 15, X = 130)}{\mathbf{P}(X = 130)} \\
 &= \frac{0.42}{0.70} = 0.60
 \end{aligned}$$

$$\begin{aligned}
 p_{Y|X}(16|130) &= 1 - p_{Y|X}(15|130) \\
 &= 1 - 0.60 \\
 &= 0.40
 \end{aligned}$$

# Continuous Conditional Distributions

- Let  $X$  and  $Y$  be jointly **continuous** random variables, with joint probability density function  $f_{X,Y}(x,y)$ , let  $f_X(x)$  denote the marginal density function of  $X$  and let  $x$  be any number for which  $f_X(x) > 0$ .
  - The conditional probability function of  $Y$  given  $X = x$  is

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

## Continuing Example 2.54

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{6}(x+y) & \text{if } 1 \leq x \leq 2 \text{ and } 4 \leq y \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- Marginal PDFs

$$f_X(x) = \frac{1}{6}(x + 4.5) \quad \text{for } 1 \leq x \leq 2 \quad f_Y(y) = \frac{1}{6}(y + 1.5) \quad \text{for } 4 \leq y \leq 5$$

- Example 2.61:
  1. Find the conditional probability density function of  $Y$  given  $X=1.2$ .
  2. Find the probability that the hole diameter is less than or equal to 4.8 mm given that the thickness is 1.2 mm.

# Answer

- Marginal PDFs

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{6}(x+y) & \text{if } 1 \leq x \leq 2 \text{ and } 4 \leq y \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

$$f_X(x) = \frac{1}{6}(x + 4.5) \quad \text{for } 1 \leq x \leq 2 \quad f_Y(y) = \frac{1}{6}(y + 1.5) \quad \text{for } 4 \leq y \leq 5$$

1.

$$\begin{aligned} f_{Y|X}(y|1.2) &= \frac{f(1.2, y)}{f_X(1.2)} \\ &= \begin{cases} \frac{1/6(1.2+y)}{1/6(1.2+4.5)} & \text{if } 4 \leq y \leq 5 \\ 0 & \text{otherwise} \end{cases} \\ &= \begin{cases} \frac{(1.2+y)}{5.7} & \text{if } 4 \leq y \leq 5 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

2.

$$\begin{aligned} \mathbf{P}(Y \leq 4.8 | X = 1.2) &= \int_{-\infty}^{4.8} f_{Y|X}(y|1.2) dy \\ &= \int_4^{4.8} \frac{1.2+y}{5.7} dy \\ &= 0.786 \end{aligned}$$

# Conditional Expectation

- Expectation is another term for mean
- A **conditional expectation** is an expectation, or mean, calculated using the conditional probability mass function or conditional probability density function
- The conditional expectation of  $Y$  given  $X = x$  is denoted by  $\mathbf{E}(Y|X = x)$  or  $\mu_{Y|X=x}$



# More on Conditional Expectation

- Let  $X$  and  $Y$  be two random variables associated with the same experiment.
  - The conditional expectation of  $X$  given event  $A$  with  $\mathbf{P}(A) > 0$ , is defined by

$$\mathbf{E}[X|A] = \sum_x x p_{X|A}(x)$$

$$\mathbf{E}[X|A] = \int_{-\infty}^{\infty} x f_{X|A}(x) dx$$

- For a function  $g(X)$ , it is given by

$$\mathbf{E}[g(X)|A] = \sum_x g(x) p_{X|A}(x)$$

$$\mathbf{E}[g(X)|A] = \int_{-\infty}^{\infty} g(x) f_{X|A}(x) dx$$

# Total Expectation Theorem (1/2)

- The conditional expectation of  $X$  given a value  $y$  of  $Y$  is defined by

$$\mathbf{E}[X|Y = y] = \sum_x x p_{X|Y}(x|y) \quad \mathbf{E}[X|Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx$$

– We have

$$\mathbf{E}[X] = \sum_y p_Y(y) \mathbf{E}[X|Y = y] \quad \mathbf{E}[X] = \int_{-\infty}^{\infty} \mathbf{E}[X|Y = y] f_Y(y) dy$$

- Let  $A_1, \dots, A_n$  be disjoint events that form a partition of sample space, and assume that  $P(A_i) > 0$ , for all  $i$ .

Then,

$$\mathbf{E}[X] = \sum_{i=1}^n \mathbf{P}(A_i) \mathbf{E}[X|A_i]$$

$$\mathbf{E}[g(X)] = \sum_{i=1}^n \mathbf{P}(A_i) \mathbf{E}[g(X)|A_i]$$

# Total Expectation Theorem (2/2)

- Let  $A_1, \dots, A_n$  be disjoint events that form a partition of the sample space, and for any event  $B$  with  $P(A_i \cap B) > 0$ , for all  $i$ . Then,

$$\mathbf{E}[X|B] = \sum_{i=1}^n \mathbf{P}(A_i|B) \mathbf{E}[X|A_i \cap B]$$

- Verification of total expectation theorem

$$\begin{aligned} \mathbf{E}[X] &= \sum_x x p_X(x) = \sum_x x \sum_y p_{X,Y}(x, y) \\ &= \sum_x x \sum_y p_Y(y) p_{X|Y}(x|y) \\ &= \sum_y p_Y(y) \sum_x x p_{X|Y}(x|y) \\ &= \sum_y p_Y(y) \mathbf{E}[X|Y = y] \end{aligned}$$

# Example (1 / 4)

- Chunk will go shopping for probability books for  $K$  hours. Here,  $K$  is a random variable and is equally likely to be 1, 2, 3, or 4. The number of books  $N$  that he buys is random and depends on how long he shops. We are told that

$$p_{N|K}(n|k) = \frac{1}{k}, \quad \text{for } n = 1, \dots, k.$$

- a. Find the joint PMF of  $K$  and  $N$ .
- b. Find the marginal PMF of  $N$ .
- c. Find the conditional PMF of  $K$  given that  $N = 2$ .
- d. We are now told that he bought at least 2 but no more than 3 books. Find the conditional mean and variance of  $K$ , given this piece of information.

# Example

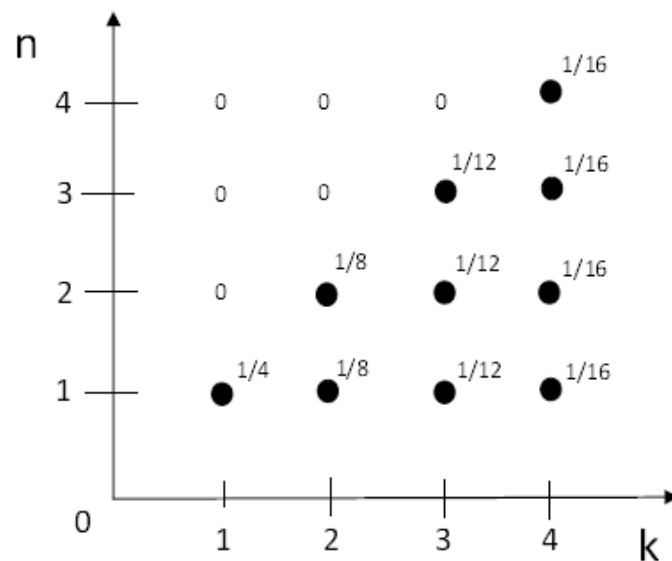
- Solutions

– a.

$$p_K(k) = \begin{cases} 1/4, & \text{for } k = 1, \dots, 4; \\ 0, & \text{otherwise} \end{cases} \quad p_{N|K}(n|k) = \begin{cases} 1/k, & \text{for } n = 1, \dots, k; \\ 0, & \text{otherwise} \end{cases}$$

$p_{N,K}(n, k) = p_{N|K}(n|k) \cdot p_K(k)$  to arrive at the following joint PMF :

$$p_{N,K}(n, k) = \begin{cases} 1/4(k), & \text{if } k = 1, 2, 3, 4 \text{ and } n = 1, \dots, k; \\ 0, & \text{otherwise} \end{cases}$$



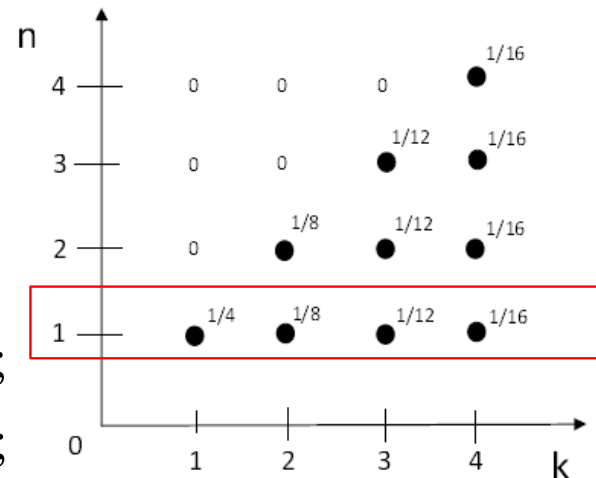
# Example

- Solutions

- b.

$$p_N(n) = \sum_k p_{N,K}(n, k) = \sum_{k=n}^4 \frac{1}{4k}$$

$$p_N(n) = \begin{cases} 1/4 + 1/8 + 1/12 + 1/16 = 25/48, & n = 1; \\ 1/8 + 1/12 + 1/16 = 13/48, & n = 2; \\ 1/12 + 1/16 = 7/48, & n = 3; \\ 1/16 = 3/48, & n = 4; \\ 0, & \text{otherwise} \end{cases}$$



- c.

$$p_{K|N}(k|2) = \frac{p_{N,K}(2, k)}{p_N(2)} = \begin{cases} 6/13, & k = 2; \\ 4/13, & k = 3; \\ 3/13, & k = 4; \\ 0, & \text{otherwise} \end{cases}$$

# Example

- Solutions

- d. Let  $A$  be the event  $2 \leq N \leq 3$ .

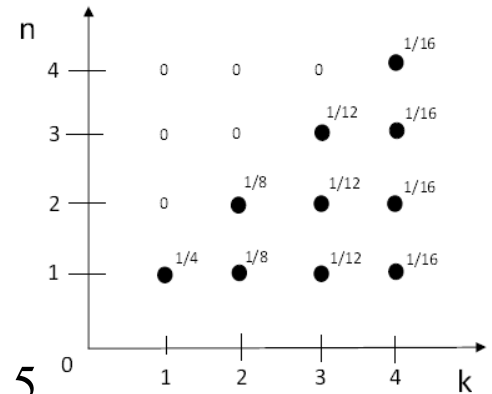
We first find the conditional PMF of  $K$  given  $A$ .

$$p_{K|A}(k) = \frac{\mathbf{P}(K = k, A)}{\mathbf{P}(A)} \quad \mathbf{P}(A) = p_N(2) + p_N(3) = \frac{5}{12}$$

$$\mathbf{P}(K = k, A) = \begin{cases} \frac{1}{8}, & k = 2; \\ \frac{1}{12} + \frac{1}{12}, & k = 3; \\ \frac{1}{16} + \frac{1}{16}, & k = 4; \\ 0, & \text{otherwise} \end{cases} \quad p_{K|A}(k) = \begin{cases} \frac{3}{10}, & k = 2; \\ \frac{2}{5}, & k = 3; \\ \frac{3}{10}, & k = 4; \\ 0, & \text{otherwise} \end{cases}$$

$$\mathbf{E}[K|A] = \sum_k k \cdot p_{K|A}(k) = 3 \quad \text{var}(K|A) = \mathbf{E}[(K - \mathbf{E}[K|A])^2 | A]$$

$$= \frac{3}{10} \cdot (2 - 3)^2 + \frac{2}{5} (3 - 3)^2 + \frac{3}{10} (4 - 3)^2 = \frac{3}{5}$$



**MORE ON INDEPENDENCE**



# Two Independent Random Variables (1/2)

- Two random variables  $X$  and  $Y$  are **independent** if

$$p_{X,Y}(x,y) = p_X(x)p_Y(y), \quad \text{for all } x,y$$

$$\text{or } \mathbf{P}(X=x \text{ and } Y=y) = \mathbf{P}(X=x)\mathbf{P}(Y=y), \quad \text{for all } x,y$$

- If a random variable  $X$  is **independent of an random variable**  $Y$

$$p_{X|Y}(x|y) = p_X(x), \quad \text{for all } y \text{ with } p_Y(y) > 0 \text{ all } x$$

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$$

$$= \frac{p_X(x)p_Y(y)}{p_Y(y)}$$

$$= p_X(x), \quad \text{for all } y \text{ with } p_Y(y) > 0 \text{ all } x$$

## Two Independent Random Variables (2/2)

- Random variables  $X$  and  $Y$  are said to be **conditionally independent**, given a positive probability event  $A$ , if

$$p_{X,Y|A}(x,y) = p_{X|A}(x)p_{Y|A}(y), \quad \text{for all } x,y$$

- Or equivalently,

$$p_{X|Y,A}(x|y) = p_{X|A}(x), \quad \text{for all } y \text{ with } p_{Y|A}(y) > 0 \text{ and all } x$$

- Note here that, as in the case of events, conditional independence may not imply unconditional independence and vice versa

# Example (1 / 2)

- **Example 2.15.** Example illustrating that conditional independence may not imply unconditional independence
  - For the PMF shown, the random variables  $X$  and  $Y$  are not independent

- To show  $X$  and  $Y$  are not independent, we only have to find a pair of values  $(x, y)$  of  $X$  and  $Y$  that

$$p_{X|Y}(x|y) \neq p_X(x)$$

- For example,  $X$  and  $Y$  are not independent

$$p_{X|Y}(1|1) = 0 \neq p_X(1) = \frac{3}{20}$$

|   |      |      |      |      |
|---|------|------|------|------|
|   | 1    | 2    | 3    | 4    |
| 4 | 1/20 | 2/20 | 2/20 | 0    |
| 3 | 2/20 | 4/20 | 1/20 | 2/20 |
| 2 | 0    | 1/20 | 3/20 | 1/20 |
| 1 | 0    | 1/20 | 0    | 0    |
|   | 1    | 2    | 3    | 4    |

# Example (2/2)

- To show  $X$  and  $Y$  are not dependent, we only have to find all pair of values  $(x, y)$  of  $X$  and  $Y$  that

$$p_{X|Y}(x|y) = p_X(x)$$

- For example,  $X$  and  $Y$  are independent, conditioned on the event  $A = \{X \leq 2, Y \geq 3\}$

$$\mathbf{P}(A) = \frac{9}{20}, \quad p_{X|Y,A}(x|y) = \frac{\mathbf{P}(X = x \cap Y = y \cap A)}{\mathbf{P}(Y = y \cap A)}$$

$$p_{X|Y,A}(1|3) = \frac{2/20}{6/20} = \frac{1}{3}, \quad p_{X|A}(1) = \frac{3/20}{9/20} = \frac{1}{3}$$

$$p_{X|Y,A}(1|4) = \frac{1/20}{3/20} = \frac{1}{3}$$

$$p_{X|Y,A}(2|3) = \frac{4/20}{6/20} = \frac{2}{3}, \quad p_{X|A}(2) = \frac{6/20}{9/20} = \frac{2}{3}$$

$$p_{X|Y,A}(2|4) = \frac{2/20}{3/20} = \frac{2}{3}$$

|   |      |      |      |      |
|---|------|------|------|------|
| 4 | 1/20 | 2/20 | 2/20 | 0    |
| 3 | 2/20 | 4/20 | 1/20 | 2/20 |
| 2 | 0    | 1/20 | 3/20 | 1/20 |
| 1 | 0    | 1/20 | 0    | 0    |
|   | 1    | 2    | 3    | 4    |

# Functions of Two Independent Random Variables

- Given  $X$  and  $Y$  be two independent random variables, let  $g(X)$  and  $h(Y)$  be two functions of  $X$  and  $Y$ , respectively. Show that  $g(X)$  and  $h(Y)$  are independent.

Let  $U = g(X)$  and  $V = h(Y)$ , then

$$\begin{aligned} p_{U,V}(u,v) &= \sum_{\{(x,y)|g(x)=u, h(y)=v\}} p_{X,Y}(x,y) \\ &= \sum_{\{(x,y)|g(x)=u, h(y)=v\}} p_X(x)p_Y(y) \\ &= \sum_{\{x|g(x)=u\}} p_X(x) \sum_{\{y|h(y)=v\}} p_Y(y) \\ &= p_U(u)p_V(v) \end{aligned}$$

# Summary on Independence (1/2)

- Random variables  $X_1, \dots, X_n$  are **independent**, provided that:
  - If  $X_1, \dots, X_n$  are **jointly discrete**, the joint probability mass function is equal to the product of the marginals:

$$p(x_1, \dots, x_n) = p_{X_1}(x_1) \cdots p_{X_n}(x_n)$$

- If  $X_1, \dots, X_n$  are **jointly continuous**, the joint probability density function is equal to the product of the marginals:

$$f(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n)$$

# Summary on Independence (2/2)

- If  $X$  and  $Y$  are **independent** random variables, then:
  - If  $X$  and  $Y$  are **jointly discrete**, and  $x$  is a value for which  $p_X(x) > 0$ , then

$$p_{Y|X}(y | x) = p_Y(y)$$

- If  $X$  and  $Y$  are **jointly continuous**, and  $x$  is a value for which

$$f_{Y|X}(y | x) = f_Y(y)$$

# Means of Two Independent Random Variables

- If  $X$  and  $Y$  are **independent** random variables, then

$$\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$$

- As shown by the following calculation

$$\begin{aligned}\mathbf{E}[XY] &= \sum_x \sum_y xy \underline{p_{X,Y}(x,y)} \\ &= \sum_x \sum_y xy \underline{p_X(x)p_Y(y)} \quad \text{by independence} \\ &= \sum_x xp_X(x) \left[ \sum_y yp_Y(y) \right] \\ &= \mathbf{E}[X]\mathbf{E}[Y]\end{aligned}$$

- Similarly, if  $X$  and  $Y$  are independent random variables, then

$$\mathbf{E}[g(X)h(Y)] = \mathbf{E}[g(X)]\mathbf{E}[h(Y)]$$



# Covariance

- Let  $X$  and  $Y$  be random variables with means  $\mu_X$  and  $\mu_Y$

- The **covariance** of  $X$  and  $Y$  is

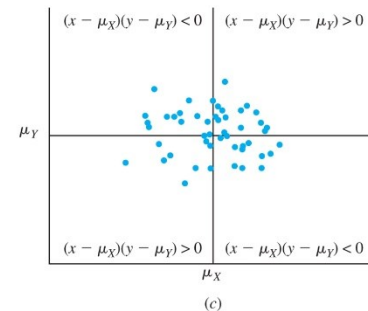
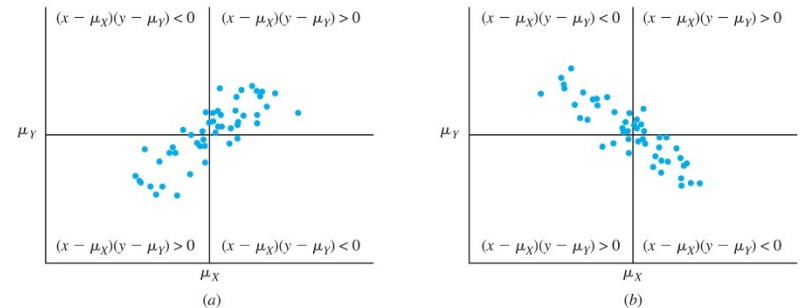
$$Cov(X, Y) = \mu_{(X - \mu_X)(Y - \mu_Y)}$$

- An alternative formula is

$$Cov(X, Y) = \mu_{XY} - \mu_X \mu_Y$$

measure of the **linear** relationship  
between  $X$  and  $Y$

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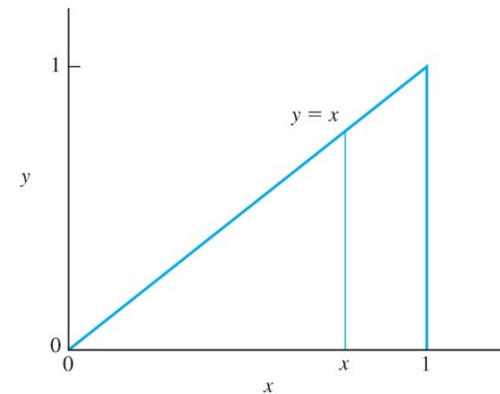


# Example

- Example 2.67: Continue Example 2.56, a mobile computer is moving in the region  $A$  bounded by the  $x$  axis, the line  $x=1$ , and the line  $y=x$ . If  $(X,Y)$  denotes the position of the computer at a given time, the joint density of  $X$  and  $Y$  is given by

$$f_{X,Y}(x,y) = \begin{cases} 8xy & (x,y) \in A \\ 0 & (x,y) \notin A \end{cases}$$

Find  $\text{Cov}(X,Y)$ .

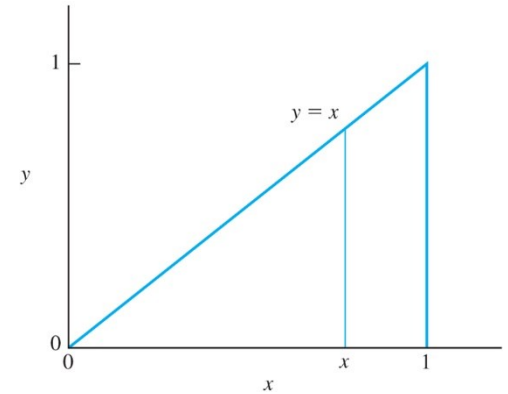


# Answer

- Example 2.67:

We will use the formula  $Cov(X, Y) = \mu_{XY} - \mu_X \mu_Y$

To compute the integral over this region, we fix a value of  $x$  as shown. We compute the inner integral by integrating with respect to  $y$  along the vertical line through  $x$ . The limits of integration along this line are  $y = 0$  to  $y = x$ . Then we compute the outer integral by integrating with respect to  $x$  over all possible values of  $x$ , so the limits of integration on the outer integral are  $x = 0$  to  $x = 1$ .



$$\begin{aligned}\mu_{XY} &= \int_0^1 \int_0^x xy(8xy) dy dx \\ &= \int_0^1 \left( \int_0^x 8x^2 y^2 dy \right) dx \\ &= \int_0^1 \frac{8x^2 y^3}{3} \Big|_{y=0}^{y=x} dx \\ &= \int_0^1 \frac{8x^5}{3} dx = \frac{8x^6}{18} \Big|_0^1 = \frac{4}{9}\end{aligned}$$

# Answer (continue)

- Example 2.67:

$$f_X(x) = \begin{cases} 4x^3 & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$
$$f_Y(y) = \begin{cases} 4y - 4y^3 & 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} \mu_X &= \int_{-\infty}^{\infty} xf_X(x)dx \\ &= \int_0^1 4x^4 dx = \left. \frac{4x^5}{5} \right|_0^1 = \frac{4}{5} \end{aligned}$$

$$\begin{aligned} \mu_Y &= \int_{-\infty}^{\infty} yf_Y(y)dy \\ &= \int_0^1 (4y^2 - 4y^4)dy = \left. \frac{4y^3}{3} - \frac{4y^5}{5} \right|_0^1 = \frac{8}{15} \end{aligned}$$

$$\text{Cov}(X, Y) = \mu_{XY} - \mu_X \mu_Y = \frac{4}{9} - \left( \frac{4}{5} \right) \left( \frac{8}{15} \right) = \frac{4}{225}$$

# Correlation

- Let  $X$  and  $Y$  be jointly distributed random variables with standard deviations  $\sigma_X$  and  $\sigma_Y$ 
  - The **correlation** between  $X$  and  $Y$  is denoted  $\rho_{X,Y}$  and is given by

$$\rho_{X,Y} = \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

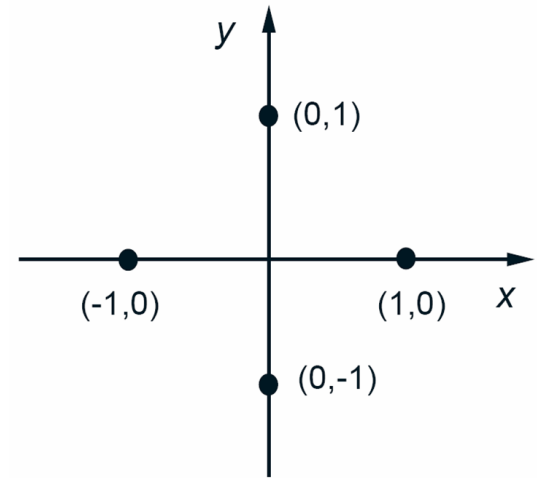
- For any two random variables  $X$  and  $Y$

$$-1 \leq \rho_{X,Y} \leq 1$$

# Covariance, Correlation, and Independence

- If  $\text{Cov}(X,Y) = \rho_{X,Y} = 0$ , then  $X$  and  $Y$  are said to be uncorrelated
- If  $X$  and  $Y$  are independent, then  $X$  and  $Y$  are uncorrelated
- It is mathematically possible for  $X$  and  $Y$  to be uncorrelated without being independent. This rarely occurs in practice

# Example



- The pair of random variables  $(X, Y)$  takes the values  $(1, 0)$ ,  $(0, 1)$ ,  $(-1, 0)$ , and  $(0, -1)$ , each with probability  $\frac{1}{4}$   
Thus, the marginal PMFs of  $X$  and  $Y$  are symmetric around 0, and  $\mathbf{E}[X] = \mathbf{E}[Y] = 0$
- Furthermore, for all possible value pairs  $(x, y)$ , either  $x$  or  $y$  is equal to 0, which implies that  $XY = 0$  and  $\mathbf{E}[XY] = 0$ .  
Therefore,  $\text{Cov}(X, Y) = \mathbf{E}[XY] - \mathbf{E}[X]\mathbf{E}[Y] = 0$ , and  $X$  and  $Y$  are **uncorrelated**
- However,  $X$  and  $Y$  are **not independent** since, for example, a nonzero value of  $X$  fixes the value of  $Y$  to zero

# Variance of a Linear Combination of Random Variables (1/2)

- If  $X_1, \dots, X_n$  are random variables and  $c_1, \dots, c_n$  are constants, then

$$\mu_{c_1X_1+\dots+c_nX_n} = c_1\mu_{X_1} + \dots + c_n\mu_{X_n}$$

$$\sigma_{c_1X_1+\dots+c_nX_n}^2 = c_1^2\sigma_{X_1}^2 + \dots + c_n^2\sigma_{X_n}^2 + 2\sum_{i=1}^{n-1}\sum_{j=i+1}^n c_i c_j \text{Cov}(X_i, X_j)$$



# Variance of a Linear Combination of Random Variables (2/2)

- If  $X_1, \dots, X_n$  are *independent* random variables and  $c_1, \dots, c_n$  are constants, then

$$\sigma_{c_1X_1+\dots+c_nX_n}^2 = c_1^2\sigma_{X_1}^2 + \dots + c_n^2\sigma_{X_n}^2$$

- In particular,

$$\sigma_{X_1+\dots+X_n}^2 = \sigma_{X_1}^2 + \dots + \sigma_{X_n}^2$$

# Summary

- Random variables: discrete and continuous
- Probability mass functions (PMF, pmf)
- Probability density functions (PDF, pdf)
- Cumulative distribution functions (CDF, cdf)
- Means and variances for random variables ( $\mu$ ,  $\sigma^2$ )
- Linear functions of random variables
- Mean and variance of a sample mean
- Jointly distributed random variables